

# Karman Singh

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## PROFESSIONAL SUMMARY

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Software Engineer and IT undergraduate at NIT Jalandhar with hands-on experience in full-stack development (MERN), cloud infrastructure (AWS, Docker, Kubernetes), and machine learning. Proven ability to design and ship production-grade systems; seeking a software engineering internship to contribute to scalable, high-impact products.

## TECHNICAL SKILLS

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**Programming Languages:** C, C++, Python, JavaScript, TypeScript, HTML, CSS, Java

**Frontend Development:** React.js, Responsive UI, REST API Integration, Component Design

**Backend Development:** Node.js, Express.js, RESTful APIs, JWT Authentication, AES-256 Encryption

**Databases:** MongoDB, SQL, Database Management Systems (DBMS)

**Cloud & DevOps:** AWS (ECS, EKS, CodePipeline, CodeBuild), Docker, Kubernetes, CI/CD Pipelines, GitHub Actions

**Core CS Fundamentals:** Data Structures & Algorithms, Operating Systems, Computer Networks, OOPs, Software Design Patterns

**Machine Learning :** Machine Learning, Computer Vision, HuggingFace Transformers, Sentiment Analysis, Remote Sensing

**Developer Tools:** Git, GitHub, Linux, NLP, Agile, SDLC

## EDUCATION

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- **Dr. B.R. Ambedkar National Institute of Technology Jalandhar, Punjab**

*Expected June 2027*

*Bachelor of Technology in Information Technology*

## WORK EXPERIENCE

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- **Summer Software Intern - IIRS, ISRO**

*June 2026 – July 2026*

- Developing a Java 21 application to automate processing of Sentinel-2 Level-2A multispectral satellite imagery for large-scale vegetation analysis.
- Implemented the MSAVI2 algorithm using Band 4 (Red) and Band 8 (NIR) and engineered a batch-processing pipeline to extract JPEG2000 spectral bands and export outputs in ENVI BSQ format.
- Gained expertise in geospatial raster processing, scientific image analysis, and remote sensing data workflows.

- **Full Stack Developer - GDGC, NIT Jalandhar**

*January 2025 - April 2026*

- Developed and maintained full-stack features for HackMol, the official hackathon platform supporting event registrations, participant management, and live hackathon operations.
- Built responsive React.js UI components and integrated RESTful APIs using Node.js and Express.js (MERN stack), improving platform user experience and reliability.
- Performed debugging and performance optimization to ensure platform stability under high-traffic conditions during live events.

## PROJECTS

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- **MindMate - AI Mental Wellness Platform** *React, Node.js, MongoDB, AWS, Docker, HuggingFace* *June 2026* 🐙

- Architected a full-stack mental wellness web app with AI-driven sentiment analysis and mood analytics using React, Node.js/Express, and MongoDB.
- Designed an asynchronous, queue-based architecture to concurrently process journal submissions and HuggingFace Transformer inference, decoupling AI workloads from the main API.
- Implemented JWT-based auth with refresh-token rotation and AES-256 field-level encryption for OWASP-compliant data privacy.
- Containerized services with Docker, deployed to AWS via GitHub Actions CI/CD pipeline, and built a real-time content moderation pipeline for an anonymous community feature.

## RELEVANT COURSEWORK

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Data Structures and Algorithms — Operating Systems — Computer Networks — Database Management Systems — Design and Analysis of Algorithms — Object-Oriented Programming — Computer Organisation and Architecture — Machine Learning

## EXTRACURRICULAR & LEADERSHIP

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- **Core Team Member — GDG, TEDxNITJalandhar & Octaves**

*NIT Jalandhar*

- Contributed to organizing technical events, workshops, and cultural programs, demonstrating leadership, teamwork, and community engagement skills.